

Simultaneous Brain Tumor Segmentation and Classification in 3D MRI with a Multi-Task UNet

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ABSTRACT

We propose a multi-task model based on the UNet architecture, capable of performing both brain tumor segmentation and classification on 3D MRI images. The model achieves an average classification accuracy of 77.07% for acoustic neuromas, meningiomas, and brain metastases, with a median Dice coefficient of 0.8069. This study demonstrates that by sharing the encoder of the UNet, the model can simultaneously perform tumor segmentation and classification tasks, highlighting the potential of a single UNet for multifunctional purposes.

Keywords: Deep learning, multi-task, brain tumor, MR Image.

1. INTRODUCTION

Convolutional neural networks (CNN) can learn to extract features and address many image recognition problems that are difficult to solve through conventional programming logic. If different task objectives can share the same neural network for image feature extraction, the

overall storage requirements on disk and the computational load during execution can be reduced. In our research, we propose a multi-task UNet model that can perform automated brain tumor segmentation and classification on 3D MRI images under limited computational resources.

In the field of medical image recognition, there have been developments of multi-task models. For instance, Huang et al.[1] proposed a method that combines pixel classification tasks with distance transform regression tasks to enhance the performance of brain tumor segmentation. He et al.[2] introduced the ACSNet model, which performs both breast tumor segmentation and classification tasks simultaneously. Additionally, Zhou et al.[3] presented a multi-task model based on the UNet architecture for tumor segmentation and classification in 3D automated breast ultrasound images. In our research, we employ a dual-pathway UNet[4] model as the backbone, with an additional classification output layer, enabling the UNet model to simultaneously perform segmentation and

classification of 3D MRI brain images.

In our study, we trained a multi-task UNet model using 3D MRI brain images from Taipei Veterans General Hospital and Taichung Veterans General Hospital. The model achieved an average classification accuracy of 77.07% for acoustic neuromas, meningiomas, and brain metastases across 458 3D test images, with a dice coefficient average of 0.7187 and a median of 0.8069. Although the Dice coefficient performance of the multi-task UNet is slightly lower than that of the single-task UNet, which has an average dice coefficient of 0.7591 and a median of 0.8407, the results of the multi-task UNet demonstrate that by sharing the same dual-channel encoder for different image recognition tasks, the CNN model can overcome the limitation of being confined to a single task.

2. MATERIALS AND METHODS

2.1. Data sets and image pre-processing

We used 3D MRI brain images from Taipei Veterans General Hospital and Taichung Veterans General Hospital as training data. This study has been approved by the Institutional Review Boards of Taipei Veterans General Hospital (IRB No.: 2021-05-05CC) and Taichung Veterans General Hospital (IRB No.: SE22478A). The study includes brain tumor types such as acoustic neuromas, meningiomas, and brain metastases. The sample sizes for the training, validation, and test sets are detailed in MODEL PREDICTION METHOD

For image patches larger than 256x256x24, we first perform 50% overlapping cropping using 256x256x24 as the standard and then predict each patch separately. In the tumor segmentation part, we average the overlapping areas of the prediction results. For the classification prediction part, we first check whether the model has segmented a tumor. If so, we retain the prediction scores for that patch;

otherwise, we discard them. Finally, we average all the. Each image includes T1-weighted contrast-enhanced and T2-weighted MRI sequences.

For each 3D MRI image, we first analyze the intensity distribution and clip intensity values below the 5th percentile and above the 95th percentile. The images are then normalized using z-score normalization. During training, we perform random rotations of ± 30 degrees along the z-axis and random left-right mirroring. Subsequently, we randomly crop regions of size 256x256x24 from the images for model training, with a batch size set to 4.

2.2. MODEL ARCHITECTURE

In this study, we adopted a dual-pathway UNet as the backbone. The encoder of this model is divided into two parts: one part uses a 3x3x1 convolutional kernel for convolutions in the xy plane, while the other part uses a 1x1x3 convolutional kernel for convolutions in the z direction. The features from both parts are then merged and input into the decoder. The purpose of dividing the encoder into two paths is to accommodate the different image resolutions in the xy and z directions in MRI images.

The encoder part employs a residual network design, similar to ResU-Net. The encoder performs two max-pooling operations in the xy plane, while the decoder performs two upsampling operations in the xy plane, keeping the z direction unchanged. The upsampling uses convolution transpose. Both the encoder and decoder use group normalization to standardize the features.

In the middle layer of the UNet, we utilize global average pooling, followed by an additional neural network branch for brain tumor classification. This branch consists of two fully connected layers and ReLU activation. The overall network architecture is shown in Figure 1.

Table 1. Total number of 3D MRI images.

Taipei Veterans General Hospital			Taichung Veterans General Hospital		
acoustic	meningiomas	metastases	acoustic	meningiomas	metastases

	neuromas			neuromas		
Train	367	869	429	90	181	550
Validation	39	100	45	8	25	68
Test	100	100	100	20	50	88

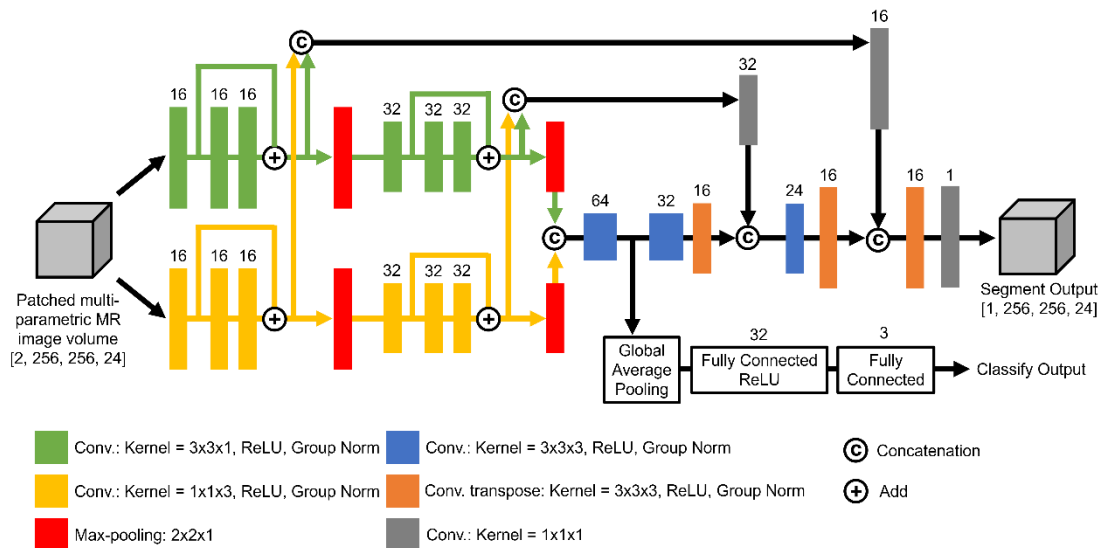


Figure 1. Multi-task UNet model architecture

2.3. LOSS FUNCTION AND OPTIMIZER

The loss functions used for training the model include the generalized dice loss[5] and cross-entropy, which are applied to tumor segmentation and classification training, respectively. These two loss functions are directly added in a 1:1 ratio. If there is no tumor in the image, only the generalized dice loss is backpropagated.

The optimizer used is AdamW, with a learning rate set to 0.001, and a weight decay of 0.01. The model is trained for a total of 200 epochs, with the learning rate multiplied by 0.95 every 10 epochs. We used mixed precision[6] training to reduce the memory usage of the computer.

2.4. MODEL PREDICTION METHOD

For image patches larger than 256x256x24, we first perform 50% overlapping cropping using 256x256x24 as the standard and then predict each patch separately. In the tumor segmentation part, we average the overlapping areas of the prediction results. For the classification prediction part, we first check whether the model has segmented a

tumor. If so, we retain the prediction scores for that patch; otherwise, we discard them. Finally, we average all the retained prediction scores to determine the overall class of the entire image.

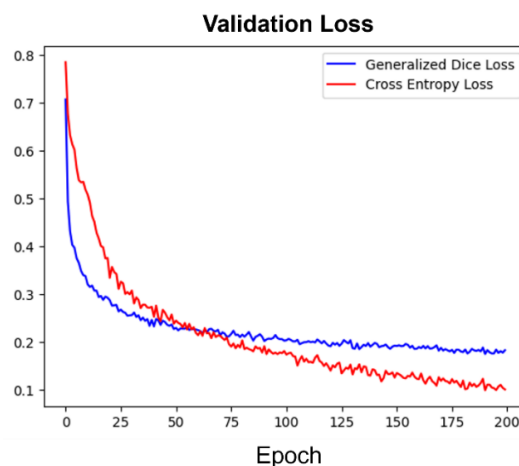


Figure 2. Validation loss curve.

3. RESULTS

Observing the validation loss during our training process, as shown in Figure 2, we can see that without special adjustment of the ratios, directly adding the generalized Dice loss and cross-entropy enables the UNet

to simultaneously train for brain tumor segmentation and classification.

We compared the Dice coefficient performance between the single-task UNet (trained solely for brain tumor segmentation) and the multi-task UNet using 458 independent 3D MRI test samples. The results, presented in Table 2, show that both models were trained under the same conditions. The results indicate that the Dice coefficient performance of the multi-task UNet is slightly lower than that of the single-task UNet. For brain tumor classification, the multi-task UNet achieved a classification accuracy of 77.07% for acoustic neuroma, meningioma, and brain metastases, with the confusion matrix shown in [錯誤! 找不到參照來源。](#).

Table 2. Dice coefficient performance of single-task UNet and multi-task UNet.

Dice Coefficient	Average	Median
Single-Task UNet	0.7591	0.8407
Multi-Task UNet	0.7187	0.8069

4. DISCUSSION

In our study, although the results indicate that the Dice coefficient for brain tumor segmentation with the multi-task UNet is approximately 0.04 lower than that of the single-task UNet model, these findings highlight the potential of the dual-channel UNet for multi-tasking (brain tumor segmentation and classification). Furthermore, there is potential for further performance enhancement of the multi-task UNet, such as by adjusting the ratio of generalized Dice loss to cross-entropy or by employing a staged training approach, where UNet is first trained on one task before learning the second task.

We also observed that during the 200 epochs of training, the validation loss shown in Figure 2 did not exhibit overfitting, indicating that adjusting the number of epochs or learning rate strategies could further improve the performance of the multi-task UNet.

Overall, our research demonstrates the potential of the dual-channel UNet in 3D MRI brain imaging to perform different tasks using a shared encoder, providing insights for future enhancements of UNet with additional functionalities.

Table 3. Multi-task UNet classification results.

		Predict		
		acoustic neuromas	meningiomas	metastases
Ground	acoustic neuromas	92	26	2
Truth	meningiomas	6	134	10
	metastases	38	23	127
Average accuracy: 77.07%				

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